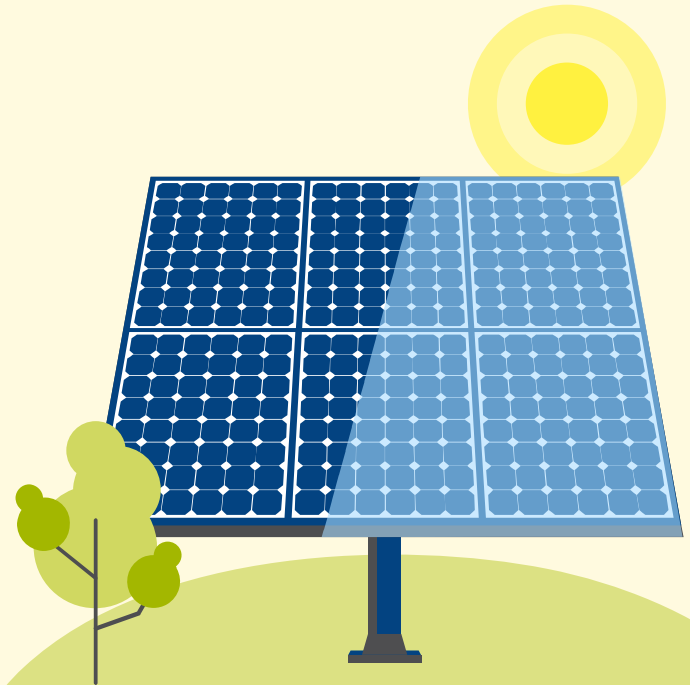


Solar Power Plant Yielding Model

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Why is this topic important?



As fossil fuels deplete, solar power is our best bet at meeting our energy needs while conserving the environment. However, due to limited accurate tracking, solar output remains unpredictable. We hope our model will solve this issue by:

- Integrating real-time atmospheric data to precisely forecast rapid shifts in solar irradiance.
- Providing reliable yield data to prevent power shortages and manage energy distribution efficiently.
- Converting unpredictable weather patterns into bankable, low-risk data that gives investors confidence.



How will the data be used and the potential impact?



- We will be training the machine learning model using data comprising of solar power generation measurements and the weather conditions they were recorded under.
- Our datasets represents various data recorded at various times under varying weather conditions from a plant from India in hopes to train the model to predict solar output fairly in all climates and times of day.
- The model will aim to make accurate predictions that can help improve energy planning, reduce operating costs, and support the transition toward renewable energy for all by reducing the risk factor.



Machine Learning Algorithms that are going to be used



Multiple Linear Regression

This algorithm will serve as the baseline to measure how independent weather variables impact the output of solar power generation. We hope it will provide a benchmark comparison against the more complex algorithm.

Logistic Regression

This model classifies whether a specific time period will produce high or low solar energy. By doing so, it helps identify peak production periods to optimize energy management.

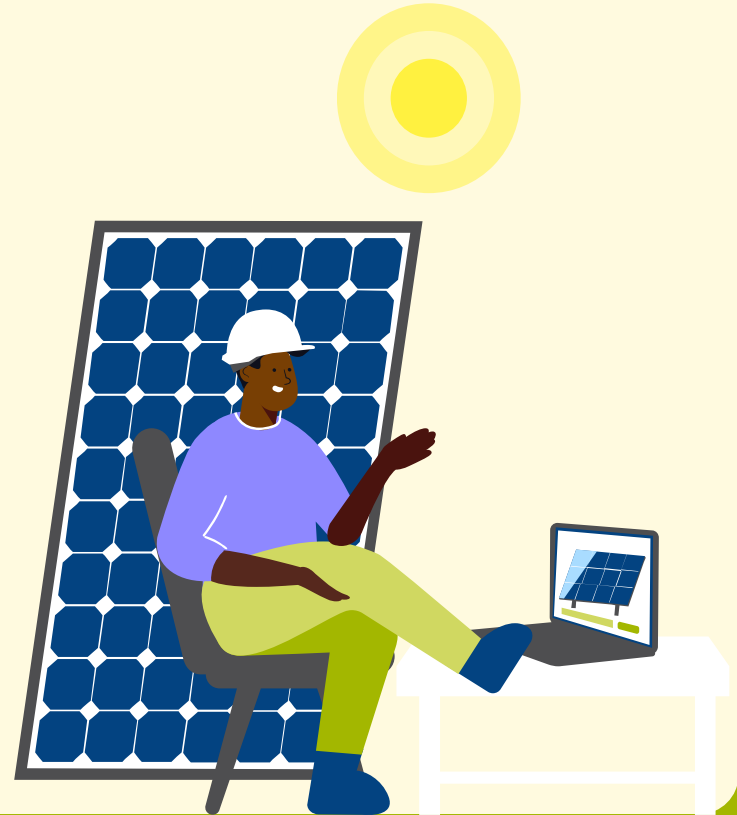
Random Forest

This will serve as our main model that will capture the complex, non-linear relations between the varying weather conditions. Ultimately this will fortify the output by weeding out data and reduce generalization of solar output.



Research Question

Can machine learning models use historical weather sensor data and solar power generation measurements from different environmental conditions to accurately predict future solar power output?





Dataset(s)



- **Name:** Solar Power generation Data - Ani Kannal
- **Source:** Kaggle
- **Link:**
https://www.kaggle.com/datasets/anikannal/solar-power-generation-data/data?select=Plant_2_Generation_Data.csv
- **Summary:** This dataset contains weather sensor and power generation data from two plants from India. It tracks 34 days of data across 20 different systems every 15 minutes for power generation and every hour for weather generation.



Training and Testing Considerations



- We should keep night time data despite it being zero to ensure the model is trained to predict to be more well rounded for seasonal shifts.
- Our dataset is limited to India and only covers dry, hot weather, so we need to be extra mindful in making sure our algorithms are accurate in tracking the independent weather conditions.
- We need to track efficiency in conversion of power system while simultaneously tracking efficiency over time.
- The dataset has to be free of null values to ensure the algorithm does not get skewed.





Sources of Bias



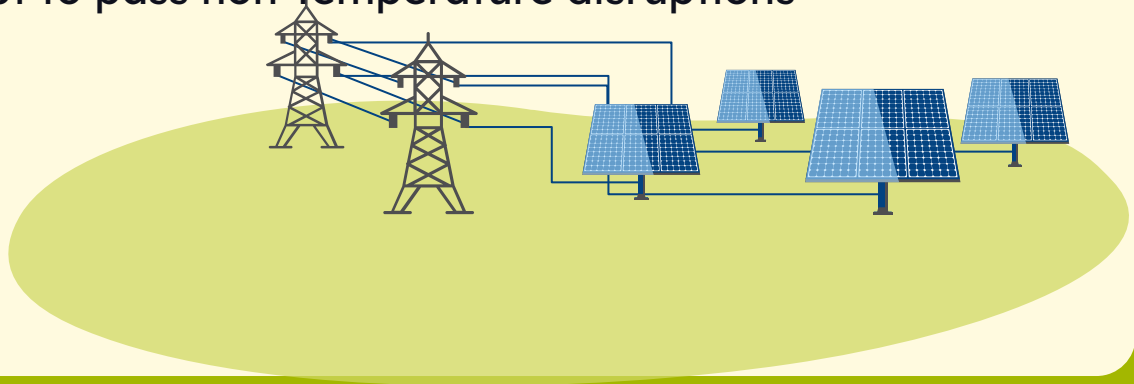
- The dataset only covers data through May and June in a 34 day period which leans more towards peak summer conditions
- Geographical bias exist as we are only limited to the tropical monsoon climate of India.
- There is insufficient representation of system and inverter which will limit the efficiency consideration.
- A lack of tracking for non meteorological reason that could interfere with the solar output.



Mitigation Strategies



- We will create multiple variables that track efficiency to create more variables to track output.
- Take out any major discrepancies in data to make sure the data is as primed to be used for training as possible.
- We will use random forest to use multiple different variables to predict the output in an attempt to pass non-temperature disruptions





Citations and Sources



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